PSS Scalp XAUUSD — Full EDA Report

Generated: 2026-11-05 · **Instrument:** XAUUSDm · **Notebooks:** `00_data_audit` → `04_eda_comparison` · **ML Engine:** XGBoost

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Step 0 — Data Audit

> 📌 **IMPORTANT**
> All numbers below are taken directly from notebook cell outputs. No values have been estimated or interpolated.

Reproducibility Anchor

Item	Value
Random seed	42
Train end	2025-05-31
Validation end	2025-11-30
Holdout start	2025-12-01
Python	3.15.2
pandas	3.1.1
numpy	2.5.4
xgboost	3.4.1

Source File Hashes (SHA-256)

File	SHA-256
`PSS_SCALP_XAUUSDm...mss h4 buy and sell,4 years.csv`	`64621afcc31f04a547de961edafad74b6976a9b9bf265544a566953fc7c56013`
`PSS_SCALP_XAUUSDm...5 years rule based.csv`	`36a1170b56b1dc6ba7e35b37bd5d93960bff1c4497863912760ad316f892e633`

Dataset Summary

Metric	4-Year (Base / ML)	5-Year (Rule-Based)
Rows read	3,045	982
Resolved trades	3,045	982
Unresolved removed	0	0
Date range	2022-01-04 15:36 → 2026-09-22 19:51	2022-01-13 01:05 → 2026-09-22 19:25
Approx span	4.71 years (1,722 days)	4.69 years (1,713 days)
Columns	91	91
Schema identical	✓ same 91 columns, same dtypes	✓
GMT offsets observed	{0.0: 3,045}	{0.0: 982}
Fully duplicated rows	0	0
Duplicated (pos_id, open_time, entry)	0	0
Entries sharing an open_time	403 (13.2%) — concurrent signals	133 (13.5%) — concurrent signals
pos_id unique within file	✓	✓

Cross-File Overlap

Check	Result
pos_id values reused across exports	505
(pos_id, open_time) collisions across files	0
(pos_id, open_time) safe as join key	✓
5-yr trades with identical entry in 4-yr	260 / 982 (26.5%)
...and identical exit as well	260
(open_time, signal) pairs in both	555 / 982
Signals in 4-yr / 5-yr	8 / 5
5-yr signals ⊆ 4-yr signals	✓

> ⚠️ **WARNING**
> The 5-year export is an **independent backtest** that reproduces a slice of the 4-year trades. The files must **not** be pooled or treated as independent samples.

Leakage Check

Post-entry / outcome columns (19) — excluded from all features:

`close_time` , `close_price` , `net_profit` , `result_R` , `pnl_R` , `exit_reason` , `MFE_R` , `MAE_R` , `min_to_MFE` , `min_to_MAE` , `min_to_1R` , `min_to_2R` , `min_to_3R` , `min_R_after_1R` , `min_R_after_2R` , `max_p_b4_sl` , `max_L_b4_tp` , `excursion_flag` , `duration_min`

Non-stationary price columns (3) — excluded: `entry` , `sl` , `tp`

Lag-feature audit (prev_R_1 / prev_R_2 / prev_R_3):

Check	4-yr	5-yr
Leaked (not realised at entry)	0 / 3,043 (0.00%)	0 / 980 (0.00%)
Matches latest closed trade	3,033 / 3,045 (99.61%)	979 / 982 (99.69%)
Residuals (earlier close used)	12 (all same sign, max | Δ | = 1.064 R)	3 (all same sign, max | Δ | = 0.003 R)
Entries with another position open	2,125 / 3,045 (69.8%)	538 / 982 (54.8%)

result_R vs pnl_R (4-yr): same sign in 100.0% of trades; |result_R| ≥ |pnl_R| in 88.1%; mean Δ = +0.0173 R → `result_R` is the cost-adjusted measure used throughout.

Feature Budget

Category	Count
Post-entry / target cols	19
Non-stationary price cols	3
ID / admin cols	4
Time cols	2
Entry-known feature cols	63
Dropped (zero information)	5 (`dist_opp_zone_R` , `planned_rr` , `risk_pct_used` , `server_gmt_offset_h` , `streak_mode`)
Usable features	**58**

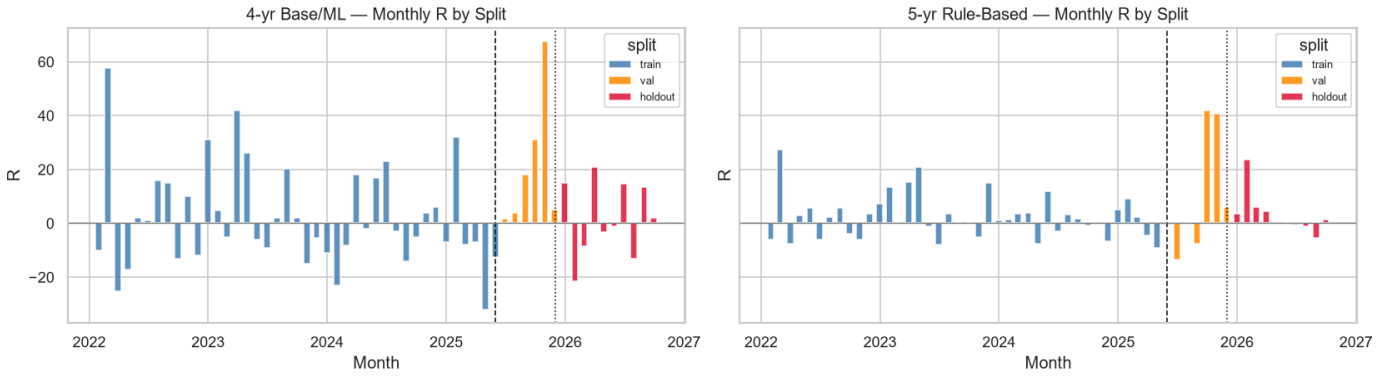
Walk-Forward Splits

4-year file (Base / ML): 5 boundary trades purged

Split	Trades	Date Range	Expectancy	Win Rate
Train	1,995	2022-01-04 → 2025-05-30	+0.0392 R	0.349
Val	496	2025-06-02 → 2025-11-28	+0.2558 R	0.419
Holdout	549	2025-12-01 → 2026-09-22	+0.0336 R	0.348

5-year file (Rule-Based): 2 boundary trades purged

Split	Trades	Date Range	Expectancy	Win Rate
Train	603	2022-01-13 → 2025-05-29	+0.1485 R	0.355
Val	186	2025-06-02 → 2025-11-27	+0.3603 R	0.425
Holdout	191	2025-12-01 → 2026-09-22	+0.1690 R	0.366



Split Overview

EDA 1 — Base Model

Hypothesis (pre-registered): The base system has a positive expectancy in R, confirmed by bootstrap, and performance is not uniform across signals, sessions and weekdays.

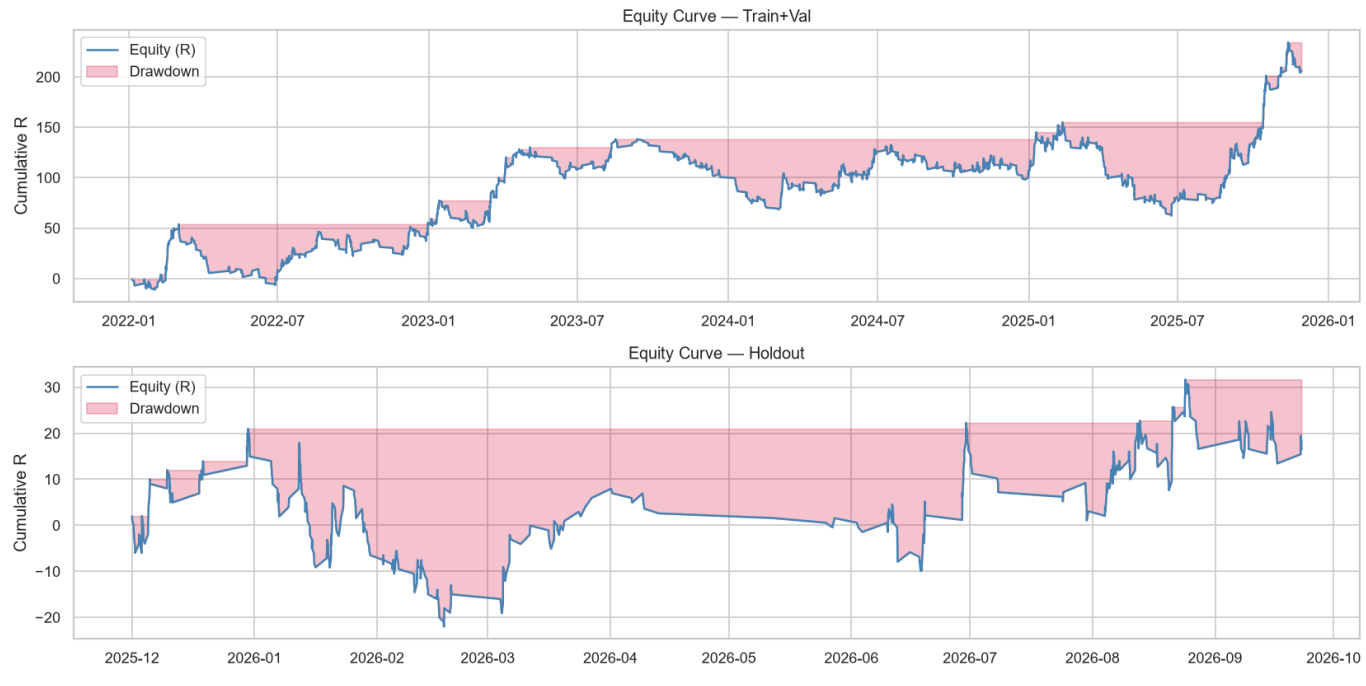
Headline Statistics

Train + Validation Period (2022-01-04 → 2025-11-28)

Metric	Value
Trades	2,491
Win rate	0.363 · 95% CI [0.344, 0.382]
Expectancy	+0.0824 R
Profit factor	1.128
Max drawdown	−92.793 R
Total R	+205.16 R

Holdout Period (2025-12-01 → 2026-09-22)

Metric	Value
Trades	549
Win rate	0.348 · 95% CI [0.309, 0.389]
Expectancy	+0.0336 R
Profit factor	1.050
Max drawdown	−42.972 R
Total R	+18.42 R



Equity Curve

Breakdowns (Train+Val)

By Signal

Signal	Trades	Win Rate	Exp R	Total R	Std R	Profit Factor
SIG_TYPE5_M30	372	0.3898	+0.1675	+62.320	1.4684	1.273
SIG_TYPE4_H1_M15	150	0.3800	+0.1395	+20.919	1.4633	1.224
SIG_BREAKER_M30	68	0.3676	+0.1026	+6.975	1.4596	1.162
SIG_TYPE5_H1_M15	197	0.3655	+0.0959	+18.898	1.4499	1.151
SIG_TYPE5_M15_M5	651	0.3641	+0.0864	+56.222	1.4609	1.134
SIG_TYPE4_M30	359	0.3593	+0.0655	+23.503	1.4716	1.100
SIG_TYPE4_M15_M3	510	0.3529	+0.0470	+23.993	1.4690	1.071
SIG_FAILEDENG_M30	184	0.3207	−0.0417	−7.673	1.4083	0.939

By Direction (Side)

Direction	Trades	Win Rate	Exp R	Total R	Std R	Profit Factor
BUY	1,581	0.3782	+0.1250	+197.606	1.4780	1.197
SELL	910	0.3363	+0.0083	+7.551	1.4246	1.012

By Session

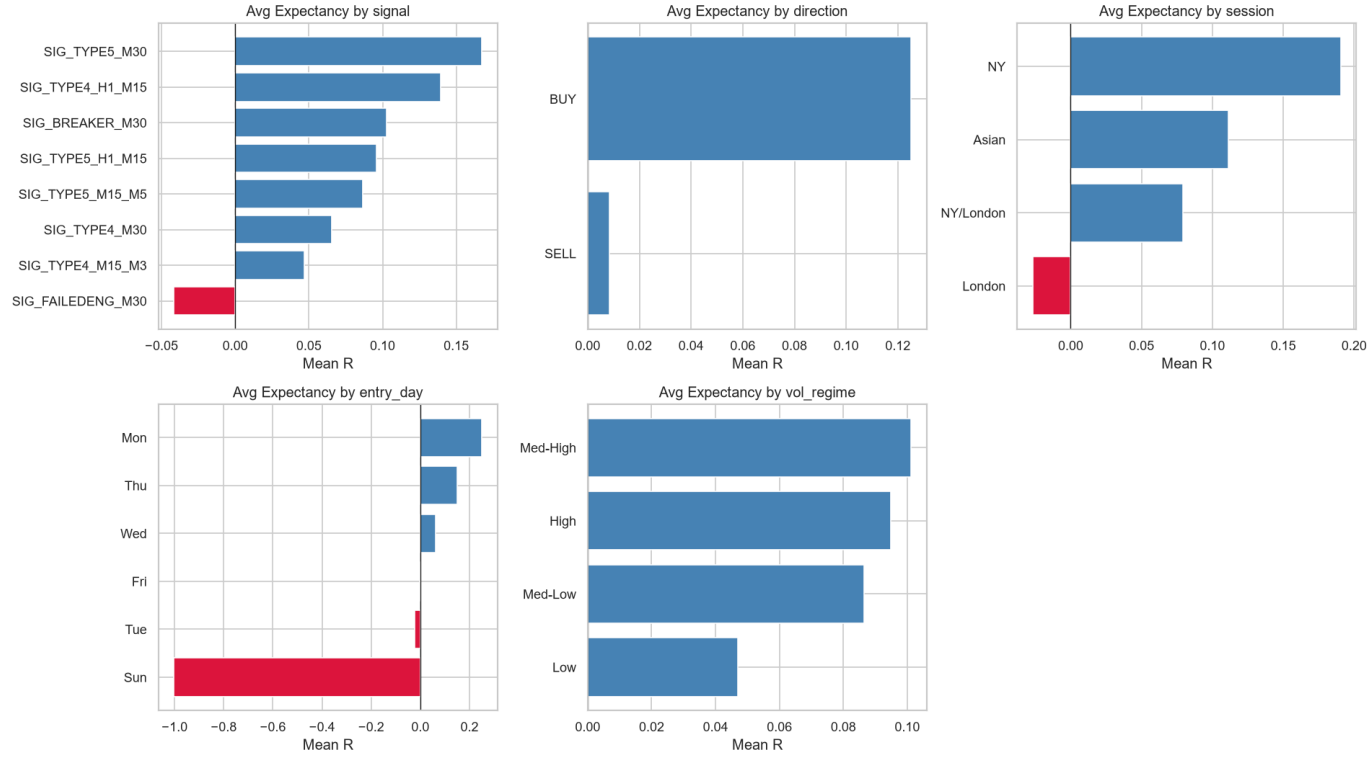
Session	Trades	Win Rate	Exp R	Total R	Std R	Profit Factor
NY	492	0.3984	+0.1906	+93.797	1.4851	1.312
Asian	741	0.3738	+0.1112	+82.387	1.4742	1.174
NY/London	592	0.3598	+0.0793	+46.924	1.4461	1.123
London	666	0.3273	−0.0270	−17.951	1.4317	0.961

By Weekday

Day	Trades	Win Rate	Exp R	Total R	Std R	Profit Factor
Mon	450	0.4156	+0.2512	+113.048	1.4897	1.429
Thu	502	0.3845	+0.1507	+75.669	1.4651	1.243
Wed	530	0.3547	+0.0606	+32.137	1.4425	1.093
Fri	480	0.3417	−0.0032	−1.524	1.4847	0.995
Tue	527	0.3264	−0.0231	−12.166	1.4121	0.966
Sun	2	0.0000	−1.0035	−2.007	0.0021	N/A

By Volatility Regime

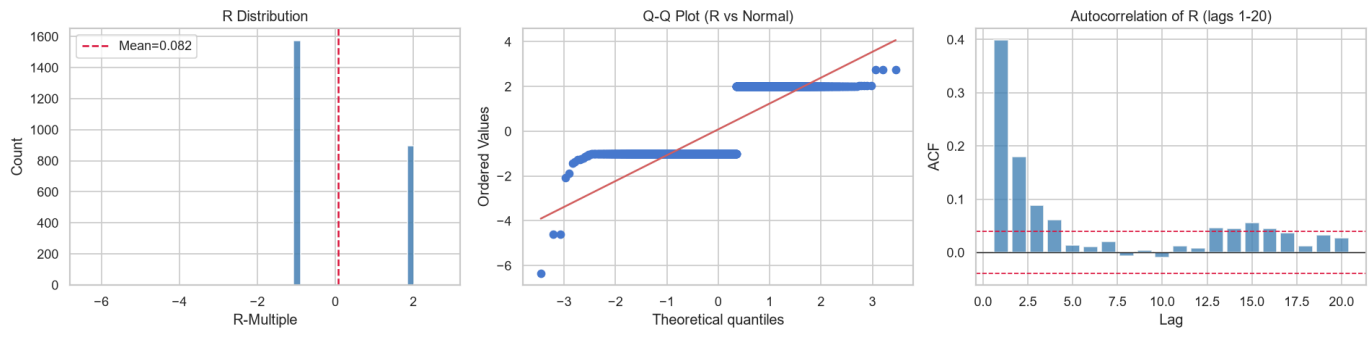
Regime	Trades	Win Rate	Exp R	Total R	Std R	Profit Factor
Med-High	621	0.3671	+0.1011	+62.772	1.4498	1.159
High	623	0.3724	+0.0948	+59.085	1.5058	1.145
Med-Low	624	0.3622	+0.0866	+54.017	1.4477	1.135
Low	623	0.3499	+0.0470	+29.283	1.4367	1.072



Breakdowns

R Distribution & Streaks

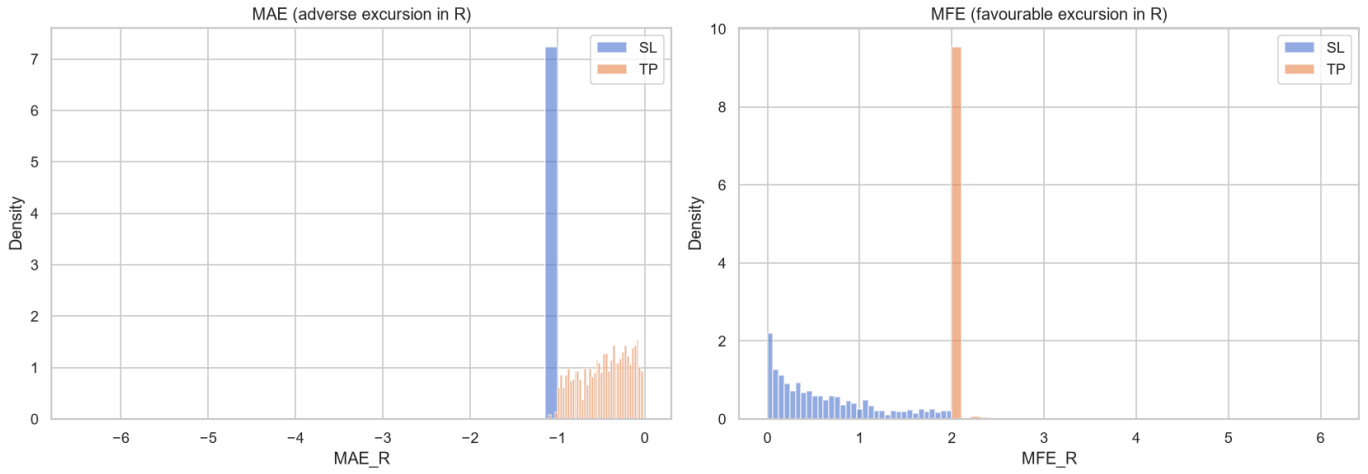
Stat	Value
Left tail (1%)	−1.013 R
Right tail (99%)	+2.009 R
Skewness	+0.507
Kurtosis	−1.467
Max losing streak — Train+Val	32
Max losing streak — Holdout	17



R Distribution

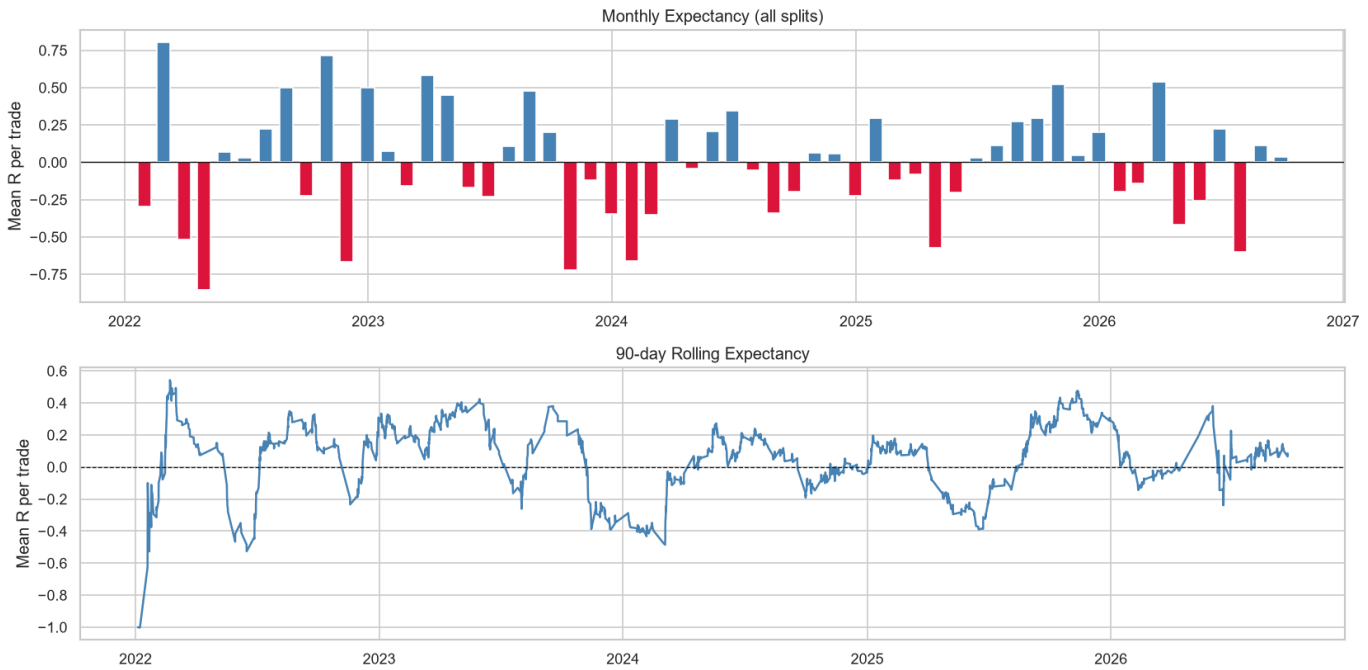
MAE / MFE by Exit Reason

Exit Reason	MAE (mean)	MFE (mean)
SL	−1.016 R	+0.613 R
TP	−0.456 R	+2.017 R



MAE / MFE

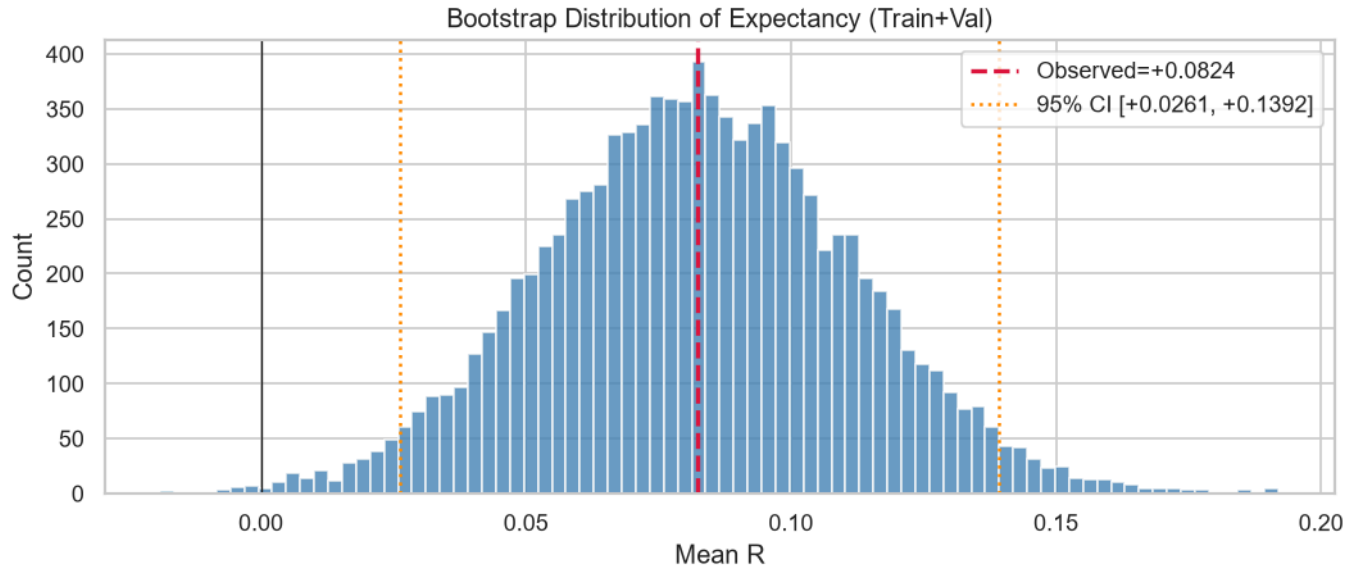
Rolling Expectancy (Monthly)



Rolling Expectancy

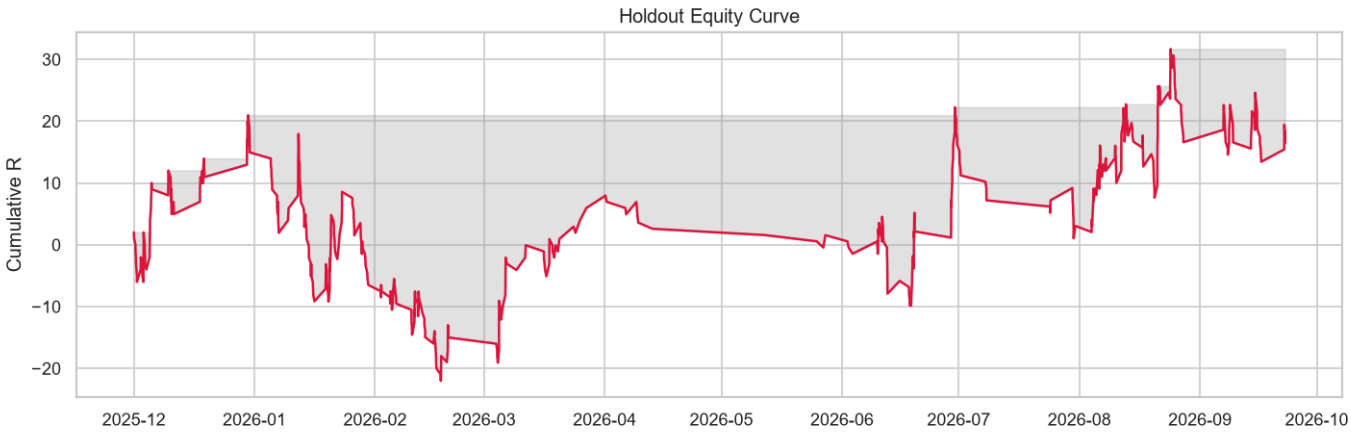
Bootstrap Edge Test (Train+Val)

Stat	Value
Observed expectancy	+0.08236 R
Bootstrap reps	10,000
95% CI	[+0.02611, +0.13923]
P(exp > 0)	0.9980



Bootstrap Expectancy

Holdout Equity Curve



Holdout Equity

EDA 2 — Rule-Based Model

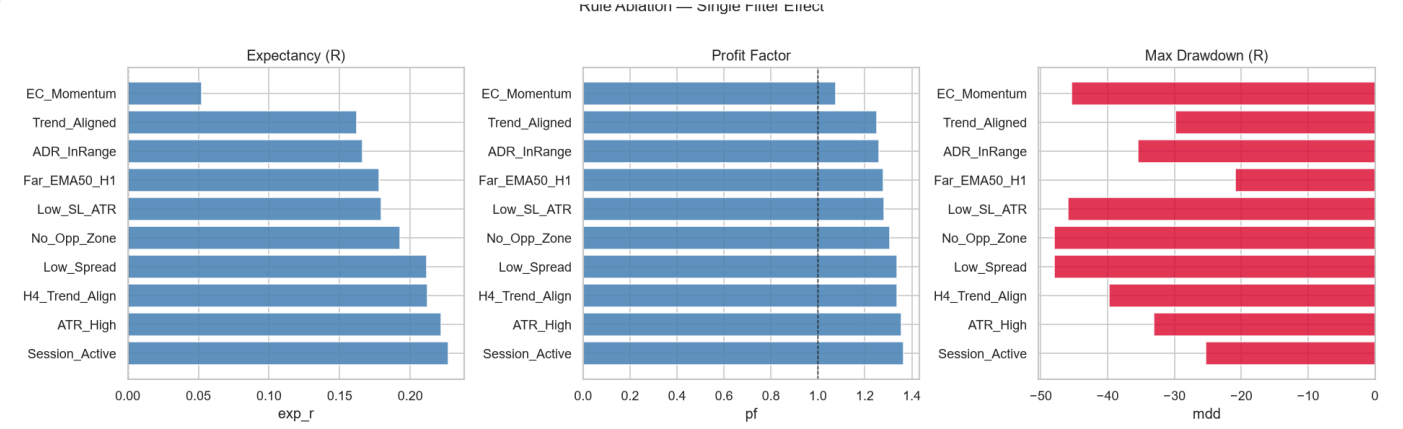
Hypothesis (pre-registered): Applying session, ATR, and trend-alignment filters improves expectancy and reduces drawdown relative to the unfiltered base, with trade frequency reduced below 20 trades/month.

Base vs Rule-Based — Full Period (2022-01-13 → 2026-09-22)

Label	N	Win Rate	Exp R	Profit Factor	Max Drawdown	Avg Trades/Month
Base (unfiltered)	3,032	0.3608	+0.0754	1.116	−92.793 R	53.2
Rule-Based	980	0.3704	+0.1927	1.305	−47.961 R	17.2

Rule Ablation (10 single-filter variants)

Filter	N	% of Trades	Win Rate	Exp R	Profit Factor	Max Drawdown	Avg Trades/Month
Session_Active	442	45.1%	0.3824	+0.2269	1.365	−25.280 R	7.8
ATR_High	781	79.7%	0.3803	+0.2217	1.356	−33.131 R	13.7
H4_Trend_Align	821	83.8%	0.3764	+0.2120	1.338	−39.744 R	14.4
Low_Spread	899	91.7%	0.3760	+0.2115	1.338	−47.961 R	16.1
No_Opp_Zone	980	100.0%	0.3704	+0.1927	1.305	−47.961 R	17.2
Low_SL_ATR	932	95.1%	0.3670	+0.1795	1.282	−45.960 R	16.4
Far_EMA50_H1	556	56.7%	0.3669	+0.1778	1.279	−20.930 R	9.8
ADR_InRange	495	50.5%	0.3616	+0.1661	1.259	−35.521 R	8.7
Trend_Aligned	706	72.0%	0.3612	+0.1621	1.252	−29.901 R	12.4
EC_Momentum	479	48.9%	0.3257	+0.0524	1.077	−45.347 R	8.6

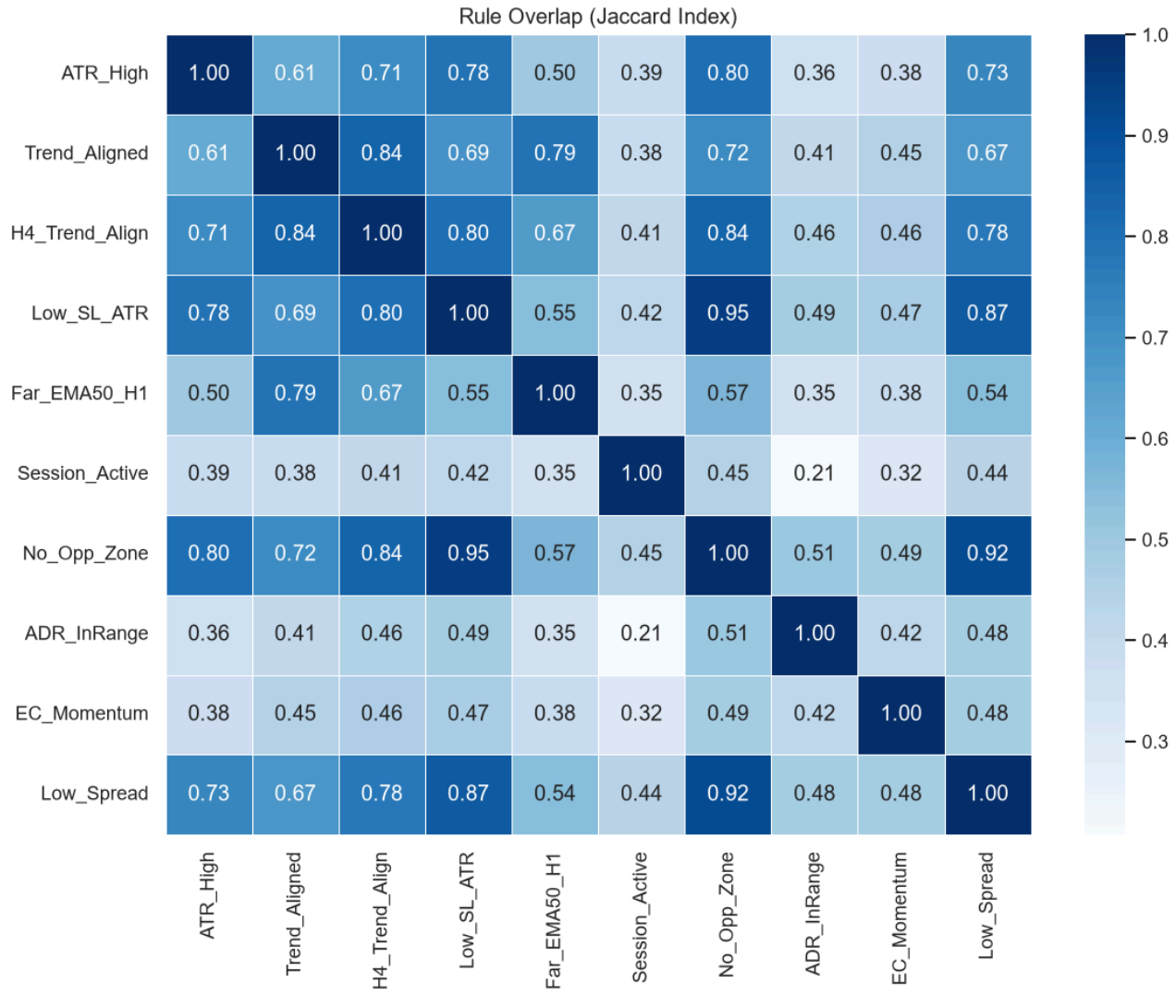


Rule Ablation

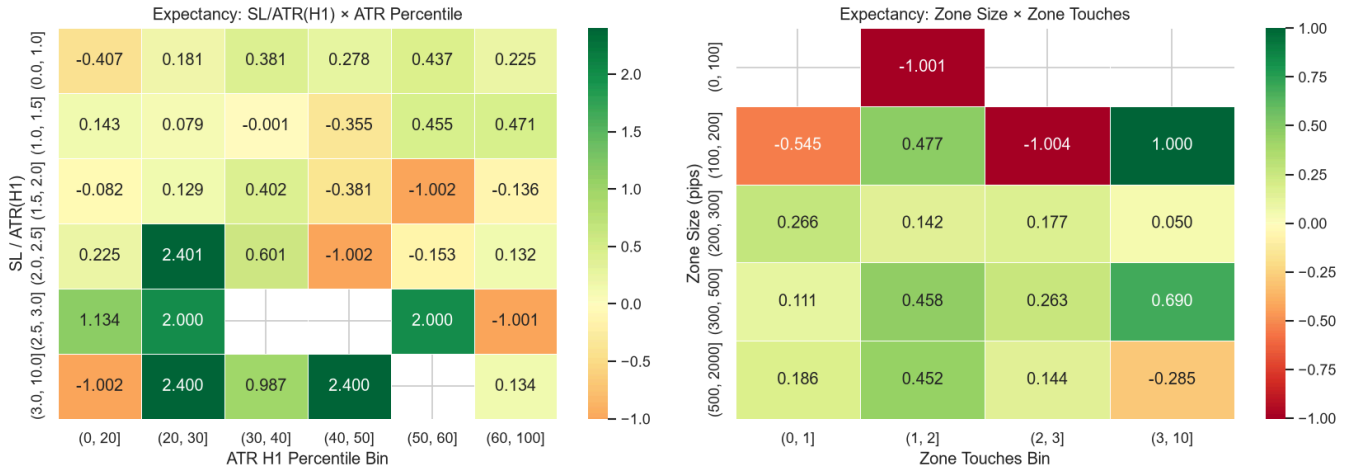
Rule Combinations (top 10 of 10 two-way combinations)

Total variants tested: 20 · Bonferroni-corrected α (0.05): 0.00250

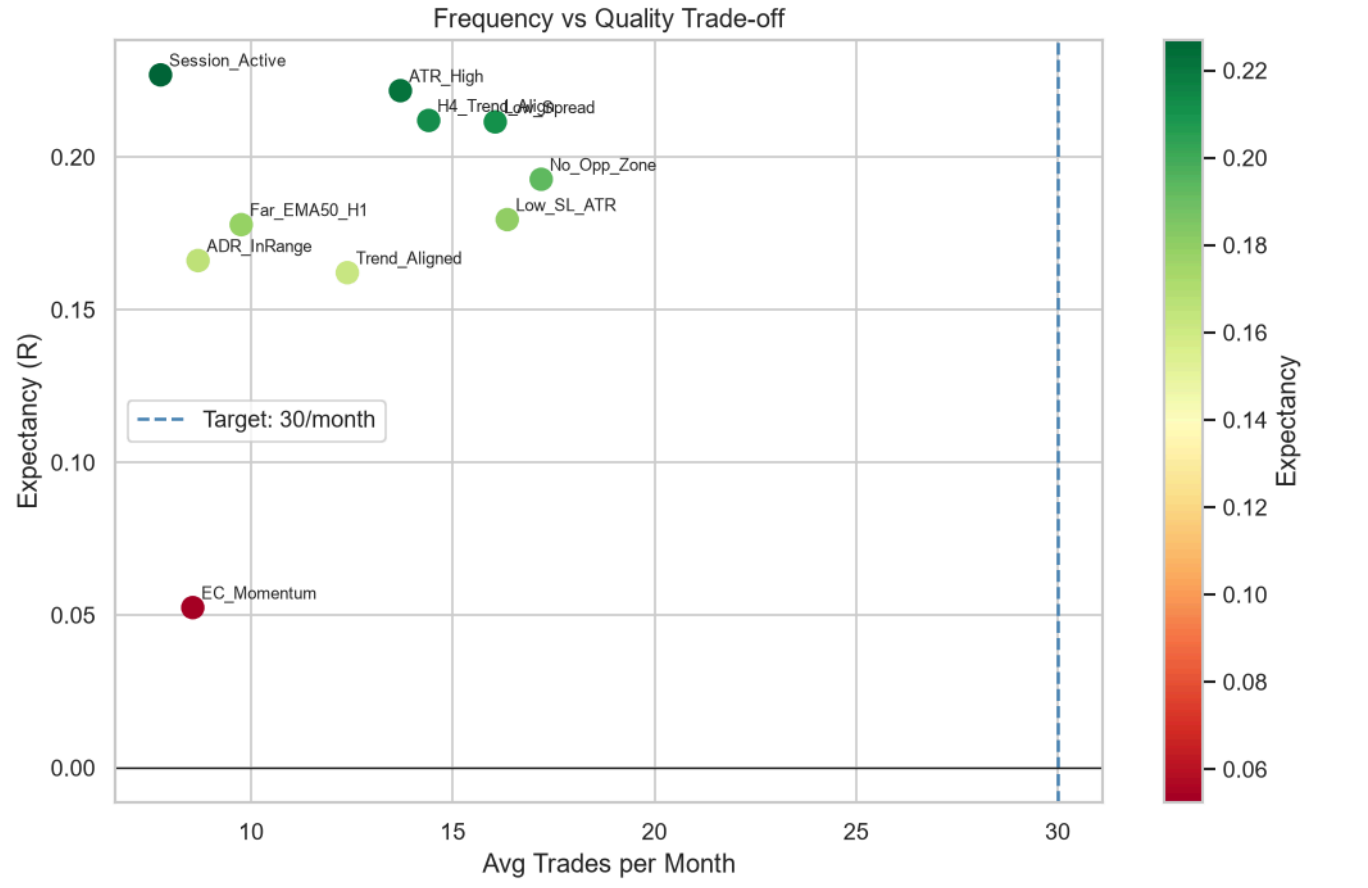
Combination	N	% Trades	Win Rate	Exp R	Profit Factor	Max Drawdown	Avg Trades/Month
Session_Active AND Low_Spread	409	41.7%	0.3961	+0.2702	1.446	−23.268 R	7.3
H4_Trend_Align AND Low_Spread	753	76.8%	0.3851	+0.2409	1.390	−39.744 R	13.4
Session_Active AND ATR_High	341	34.8%	0.3871	+0.2397	1.389	−23.254 R	6.0
ATR_High AND Low_Spread	710	72.4%	0.3831	+0.2311	1.373	−33.131 R	12.7
Session_Active AND No_Opp_Zone	442	45.1%	0.3824	+0.2269	1.365	−25.280 R	7.8
ATR_High AND No_Opp_Zone	781	79.7%	0.3803	+0.2217	1.356	−33.131 R	13.7
ATR_High AND H4_Trend_Align	666	68.0%	0.3784	+0.2150	1.344	−32.921 R	11.7
H4_Trend_Align AND No_Opp_Zone	821	83.8%	0.3764	+0.2120	1.338	−39.744 R	14.4
Low_Spread AND No_Opp_Zone	899	91.7%	0.3760	+0.2115	1.338	−47.961 R	16.1
Session_Active AND H4_Trend_Align	370	37.8%	0.3703	+0.1856	1.293	−27.066 R	6.6



Rule Overlap



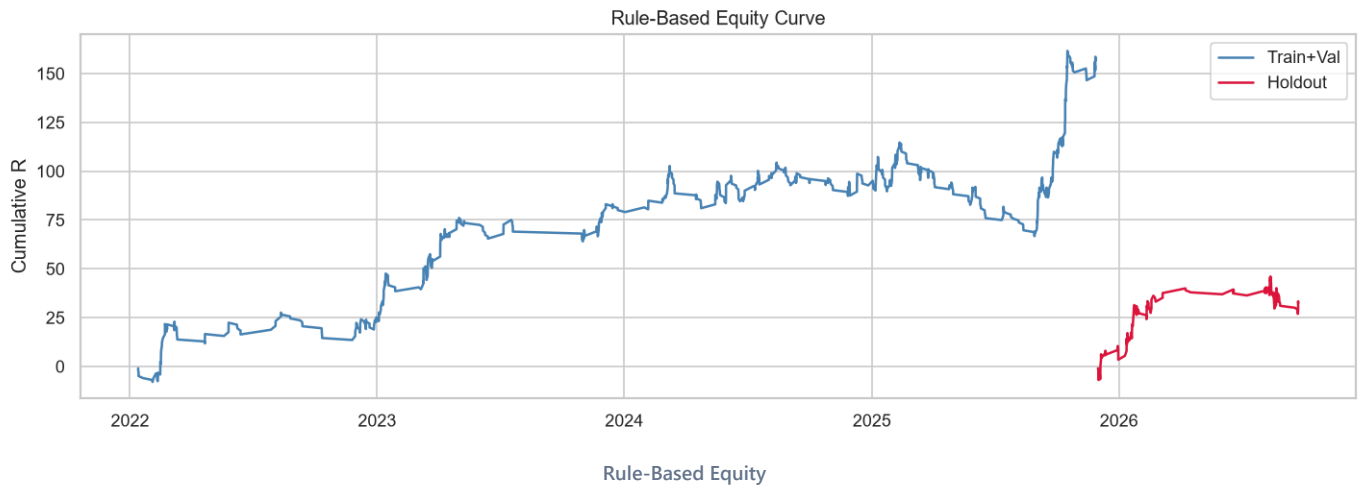
Threshold Heatmaps



Frequency vs Quality

Rule-Based — Train+Val and Holdout

Split	N	Win Rate	Exp R	Profit Factor	Max Drawdown	Avg Trades/Month
Train+Val	789	0.3714	+0.1984	1.314	−47.961 R	16.8
Holdout	191	0.3665	+0.1690	1.265	−19.062 R	19.1



Rule-Based Equity

EDA 3 — ML Variant (XGBoost)

Hypothesis (pre-registered): An XGBoost classifier trained on entry-known features provides statistically significant uplift (permutation p < 0.05) over the base model on the held-out period. Given AUC approximately 0.55 and 1,498 trades in training, signal is expected to be small.

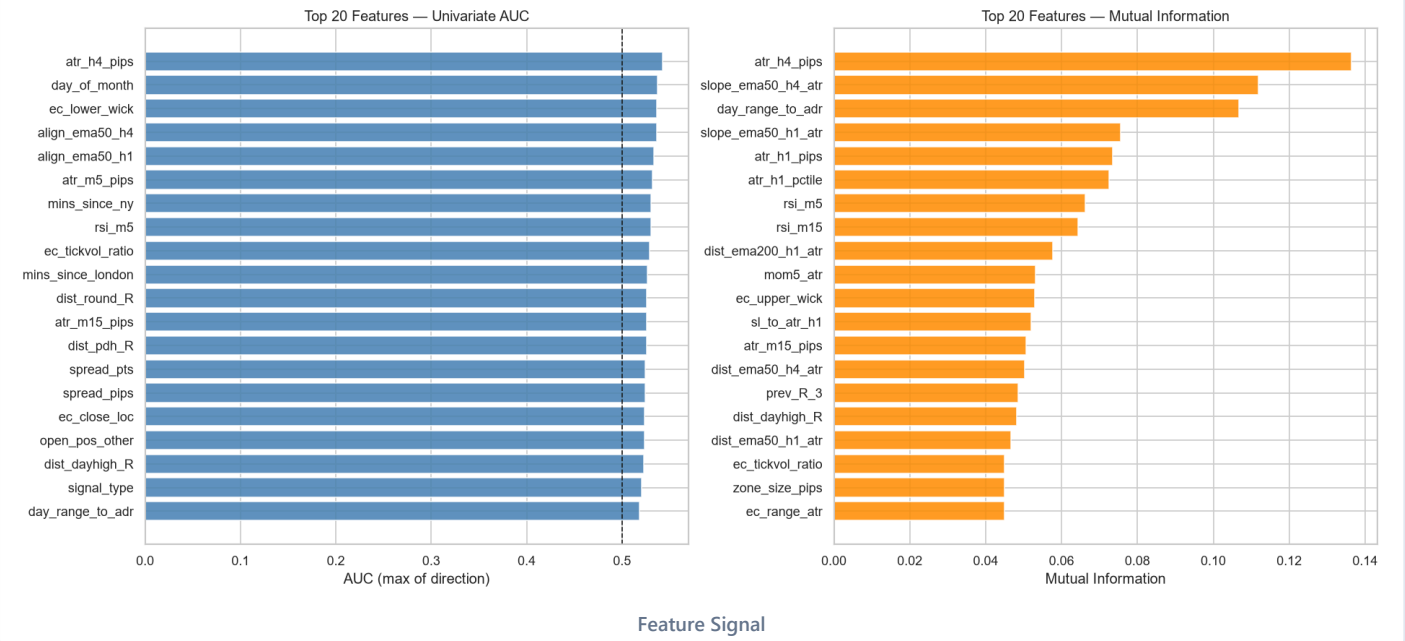
Dataset Splits Used

Split	Trades	Win Rate
Train	1,995	0.349
Val	496	0.419
Holdout	549	0.348

Feature columns used: 53

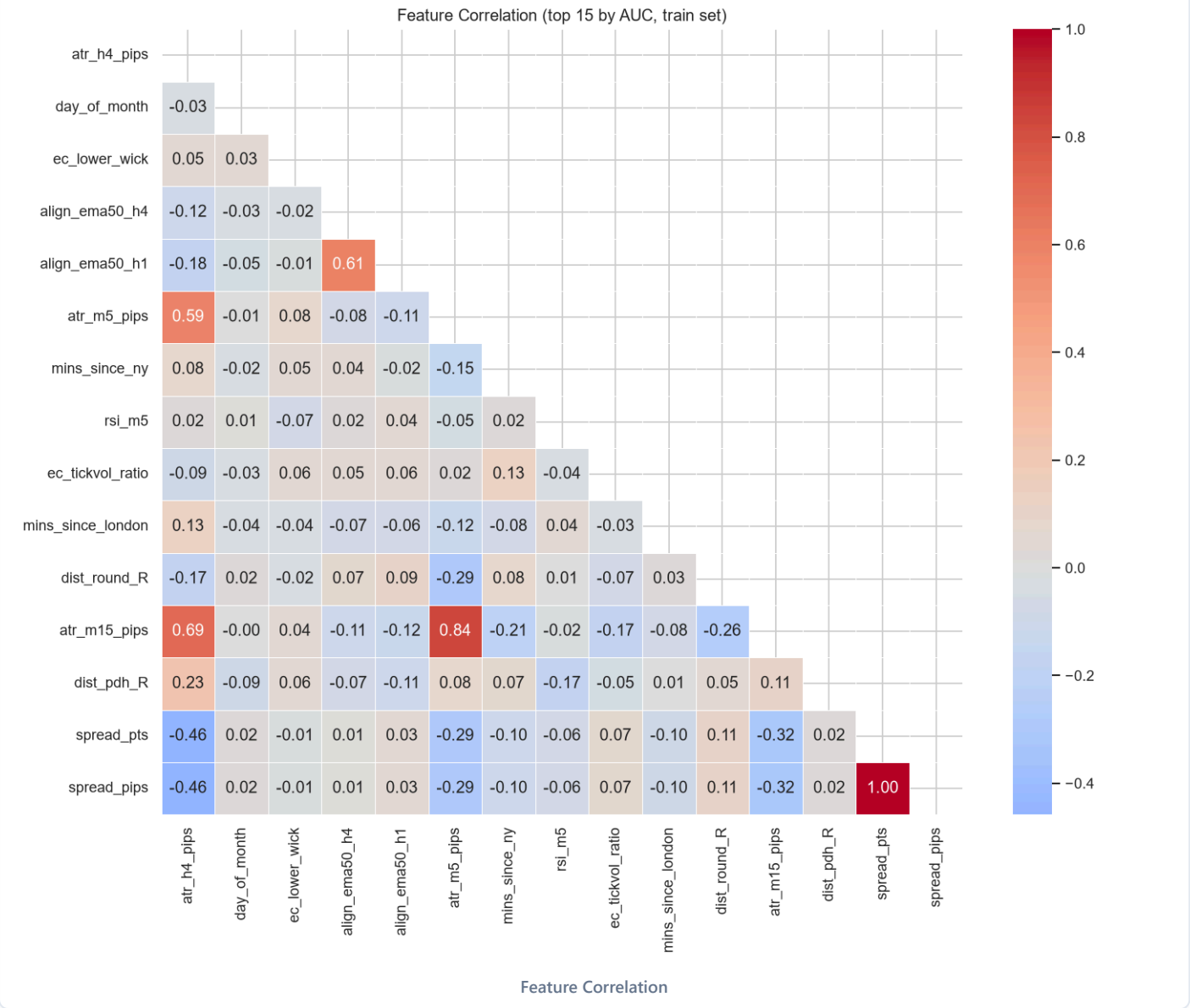
Top 20 Features by Univariate AUC

Rank	Feature	Uni AUC	Mutual Info
1	atr_h4_pips	0.5423	0.136353
2	day_of_month	0.5372	0.000000
3	ec_lower_wick	0.5366	0.016225
4	align_ema50_h4	0.5366	0.015240
5	align_ema50_h1	0.5338	0.009959
6	atr_m5_pips	0.5318	0.044063
7	mins_since_ny	0.5306	0.023346
8	rsi_m5	0.5303	0.066176
9	ec_tickvol_ratio	0.5286	0.045014
10	mins_since_london	0.5263	0.029951
11	dist_round_R	0.5258	0.044344
12	atr_m15_pips	0.5255	0.050678
13	dist_pdh_R	0.5255	0.032062
14	spread_pts	0.5240	0.001269
15	spread_pips	0.5240	0.000000
16	ec_close_loc	0.5236	0.028855
17	open_pos_other	0.5232	0.003939
18	dist_dayhigh_R	0.5228	0.048274
19	signal_type	0.5206	0.011384
20	day_range_to_adr	0.5185	0.106710



Feature Drift Check (top 5 features, mean by split)

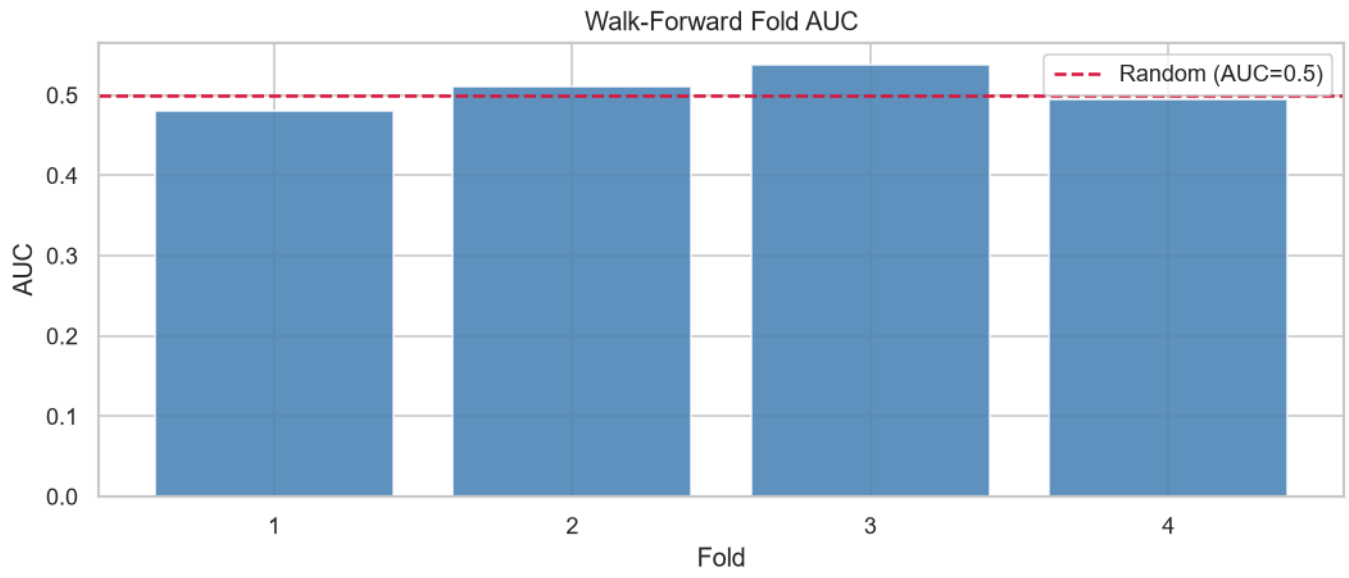
Feature	Holdout	Train	Val
atr_h4_pips	4,437.00	1,213.05	2,522.23
day_of_month	14.69	15.57	13.49
ec_lower_wick	0.1734	0.1419	0.1493
align_ema50_h4	0.5519	0.6010	0.7702
align_ema50_h1	0.3953	0.4155	0.5565



XGBoost Walk-Forward Fold AUC

Fold	Train N	Test N	AUC	Date From	Date To
1	498	498	0.4799	2023-01-05	2024-02-05
2	996	498	0.5108	2024-02-05	2024-11-07
3	1,494	498	0.5378	2024-11-08	2025-05-30
4	1,992	498	0.4950	2025-05-30	2025-11-28

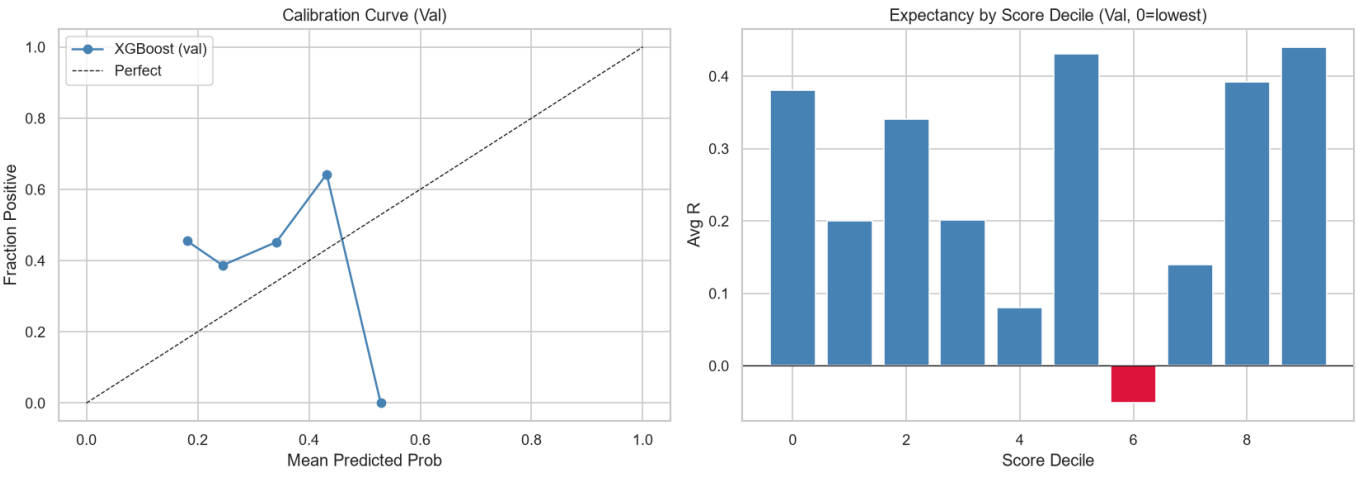
Metric	Value
Val AUC	0.5011
Holdout AUC	0.5168



Fold AUC

Calibration & Decile Lift (Val Set)

Decile	Win Rate	Exp R	N
0	0.4600	+0.3800	50
1	0.4000	+0.2000	50
2	0.4490	+0.3409	49
3	0.4000	+0.2017	50
4	0.3673	+0.0798	49
5	0.4800	+0.4303	50
6	0.3061	−0.0515	49
7	0.3800	+0.1398	50
8	0.4694	+0.3918	49
9	0.4800	+0.4402	50



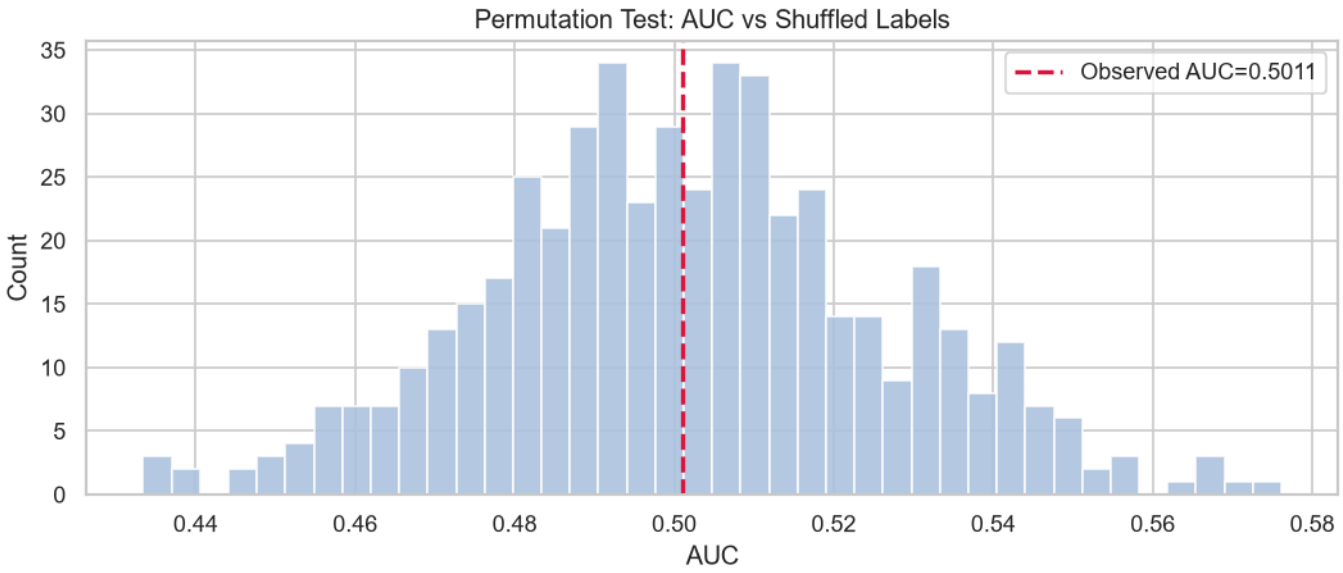
Calibration & Lift

Permutation Test (Val Set, 500 permutations)

Stat	Value
Observed AUC (val)	0.5011
Permutation AUC mean	0.5017
Permutation AUC std	±0.0250
Permutation p-value	0.5000

NOTE

With p = 0.5000, the XGBoost validation AUC is indistinguishable from random noise (shuffled labels baseline).



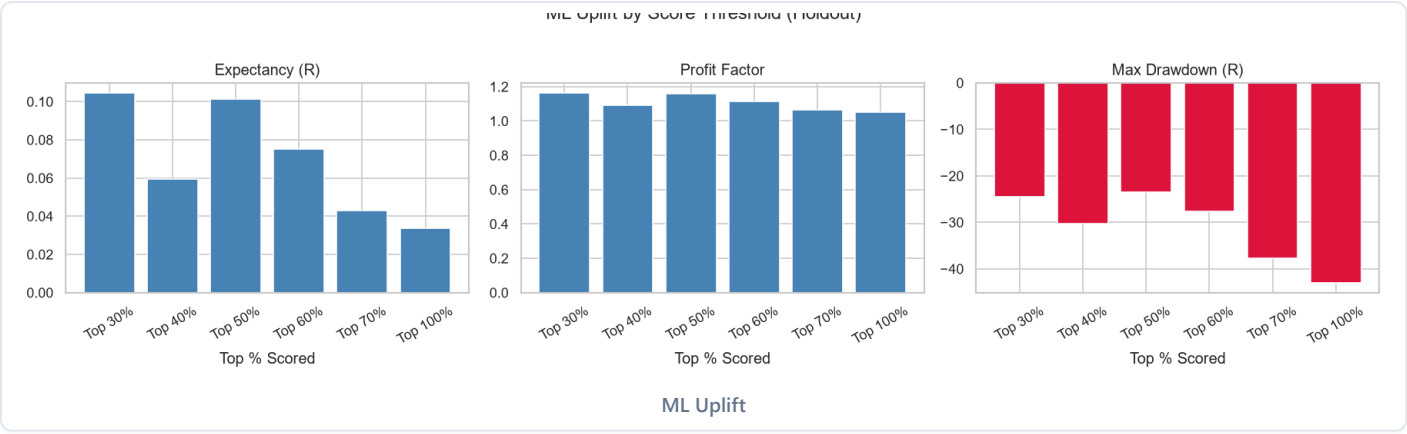
Permutation Test

ML Uplift — Holdout (top X% of scored trades)

Selection	N	Exp R	Profit Factor	Max Drawdown
Top 30%	165	+0.1046	1.164	−24.497 R
Top 40%	220	+0.0596	1.092	−30.266 R
Top 50%	275	+0.1011	1.157	−23.580 R
Top 60%	329	+0.0751	1.115	−27.666 R
Top 70%	384	+0.0429	1.065	−37.700 R
Top 100%	549	+0.0336	1.050	−42.972 R

NOTE

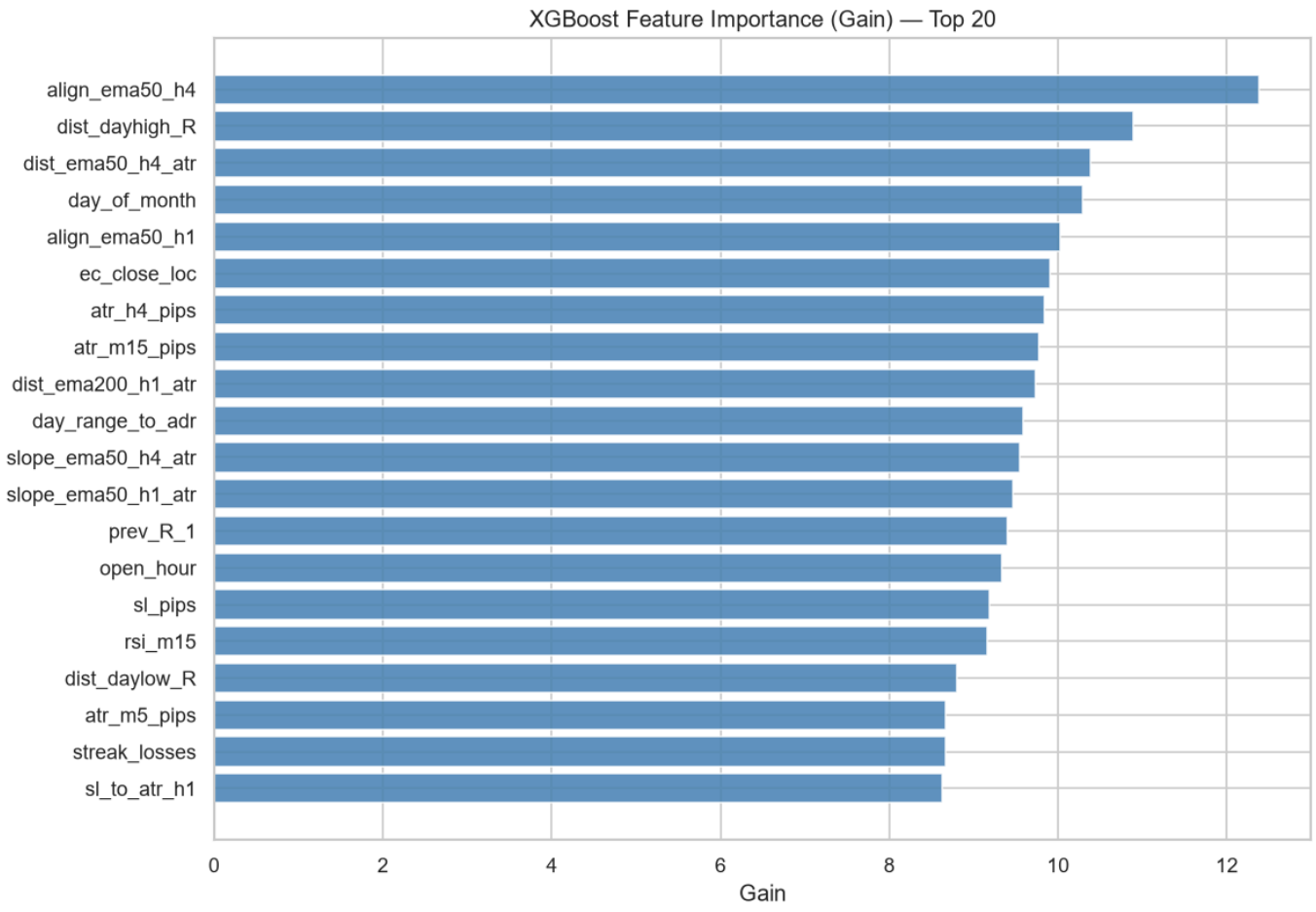
Unlike LightGBM (which suffered negative holdout uplift), XGBoost top-scored subsets produce positive uplift over the 100% baseline: Top 50% reaches +0.1011 R (PF 1.157, MDD −23.58 R) vs +0.0336 R (PF 1.050, MDD −42.97 R) for the unfiltered base.



ML Uplift

XGBoost Feature Importance (top 20 by gain)

Rank	Feature	Gain	Split Count
1	align_ema50_h4	12.380	11.0
2	dist_dayhigh_R	10.889	9.0
3	dist_ema50_h4_atr	10.383	12.0
4	day_of_month	10.291	20.0
5	align_ema50_h1	10.019	3.0
6	ec_close_loc	9.908	12.0
7	atr_h4_pips	9.839	39.0
8	atr_m15_pips	9.776	8.0
9	dist_ema200_h1_atr	9.730	13.0
10	day_range_to_adr	9.582	23.0
11	slope_ema50_h4_atr	9.546	13.0
12	slope_ema50_h1_atr	9.467	10.0
13	prev_R_1	9.400	9.0
14	open_hour	9.330	8.0
15	sl_pips	9.181	5.0
16	rsi_m15	9.163	18.0
17	dist_daylow_R	8.794	4.0
18	atr_m5_pips	8.672	15.0
19	streak_losses	8.661	5.0
20	sl_to_atr_h1	8.624	6.0



Feature Importance

EDA 4 — Comparison & Decision

Hypothesis (pre-registered): The Rule-Based model will outperform both Base and ML variants on the holdout period by expectancy and Calmar ratio, with the ML variant providing no statistically significant uplift.

Holdout Counts

Model	Holdout Trades
Base	549
Rule-Based	191
ML (top 50%)	275

Holdout Core Metrics — Base Cost

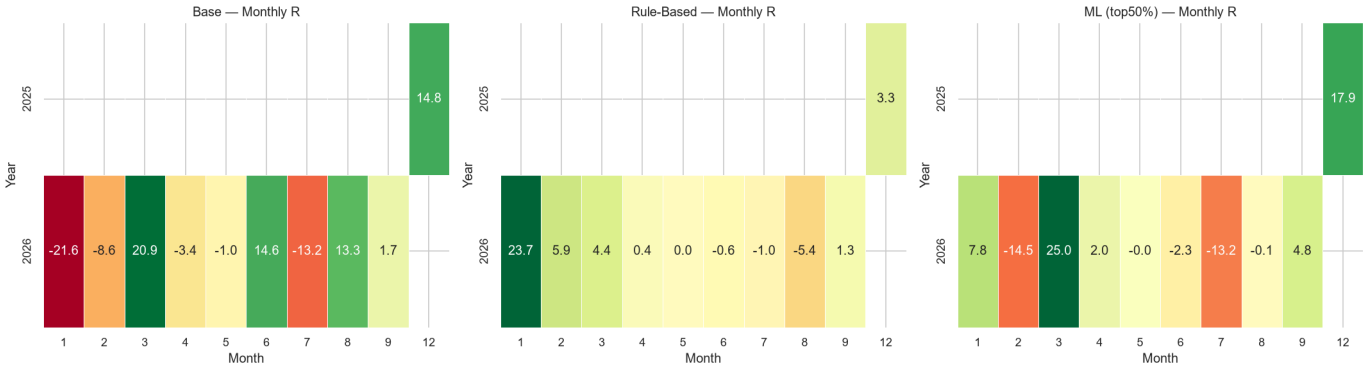
Model	N	Win Rate	Exp R	Profit Factor	Calmar	Max Drawdown	Worst Month	Max Losing Streak	Avg Trades/Month
Base	549	0.348	+0.0320	1.048	0.347	−43.212 R	−21.631 R	17	55.8
Rule-Based	191	0.366	+0.1675	1.262	1.429	−19.131 R	−5.412 R	10	19.4
ML (top 50%)	275	0.371	+0.0995	1.155	0.989	−23.627 R	−14.537 R	18	28.0

Holdout Core Metrics — Stressed Cost

Model	N	Win Rate	Exp R	Profit Factor	Calmar	Max Drawdown	Worst Month	Max Losing Streak	Avg Trades/Month
Base (stressed)	549	0.348	+0.0304	1.045	0.328	−43.451 R	−21.807 R	17	55.8
Rule-Based (stressed)	191	0.366	+0.1661	1.259	1.412	−19.200 R	−5.522 R	10	19.4
ML (top 50%) (stressed)	275	0.371	+0.0978	1.152	0.971	−23.674 R	−14.575 R	18	28.0



Equity Comparison



Monthly Heatmap

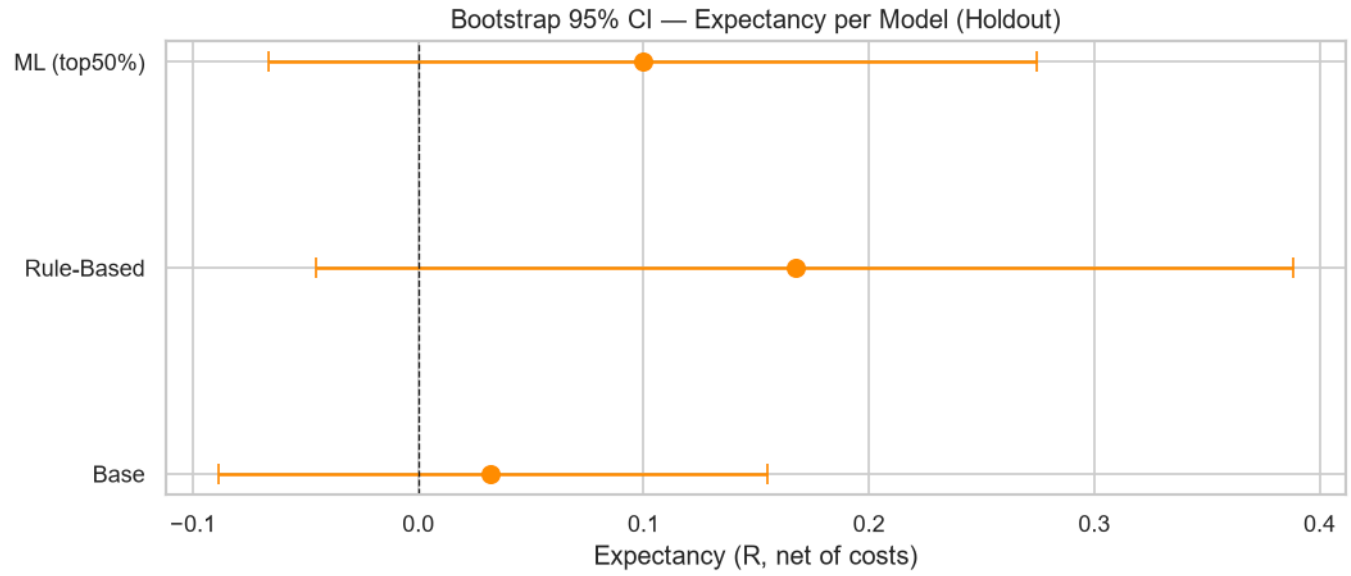
Bootstrap Significance (N = 10,000 reps, BH-corrected)

Comparison	Observed Difference	CI Lower	CI Upper	Raw p	BH-corrected p	Reject H_0
Rule-Based vs Base	+0.13558 R	−0.11213	+0.39010	0.1455	0.3220	No
ML (top 50%) vs Base	+0.06749 R	−0.14372	+0.27781	0.2635	0.3220	No
Rule-Based vs ML (top 50%)	+0.06810 R	−0.21219	+0.35770	0.3220	0.3220	No

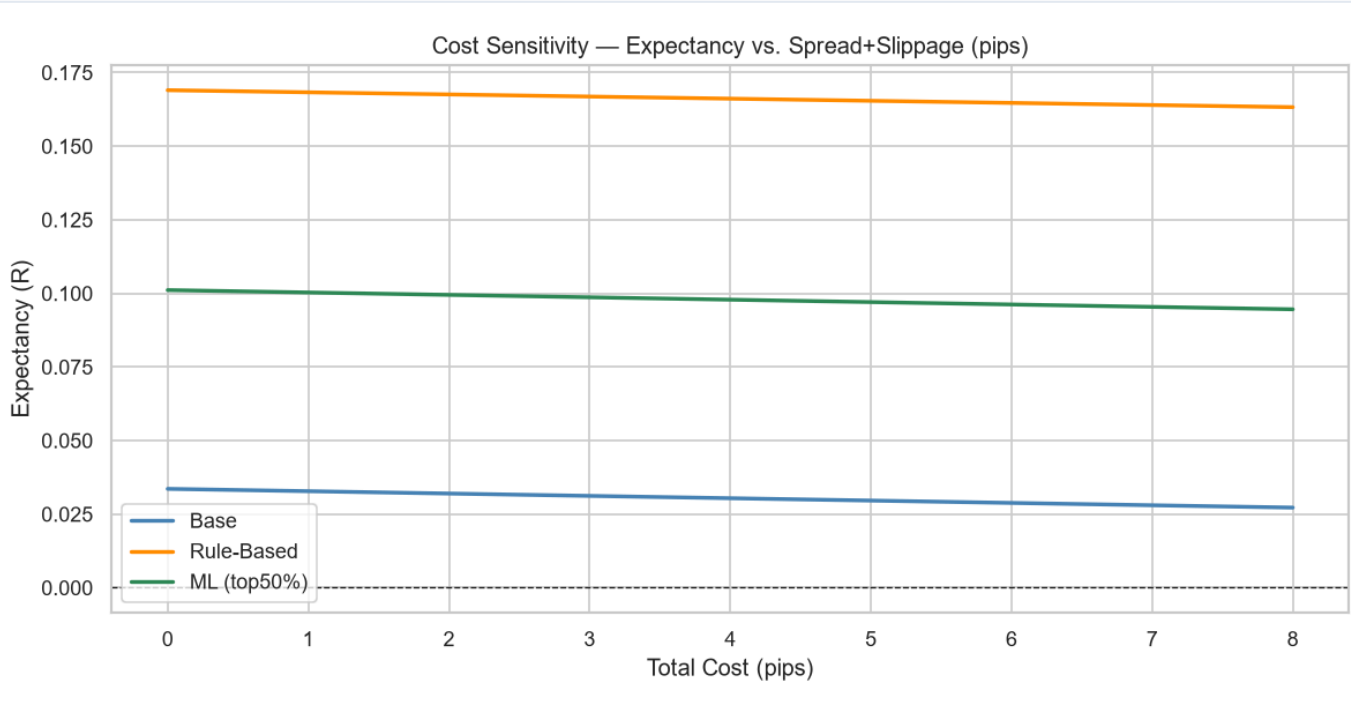
None of the pairwise comparisons reject H_0 after Benjamini-Hochberg FDR correction.

Bootstrap CI by Model — Expectancy (Holdout)

Model	Exp R (observed)	CI Lower	CI Upper	P(exp > 0)
Base	+0.03197	−0.08842	+0.15473	0.7037
Rule-Based	+0.16755	−0.04566	+0.38803	0.9296
ML (top 50%)	+0.09945	−0.06636	+0.27449	0.8730



Bootstrap CI



Cost Sensitivity

Pre-Specified Decision Rule (applied to Holdout, Base Cost)

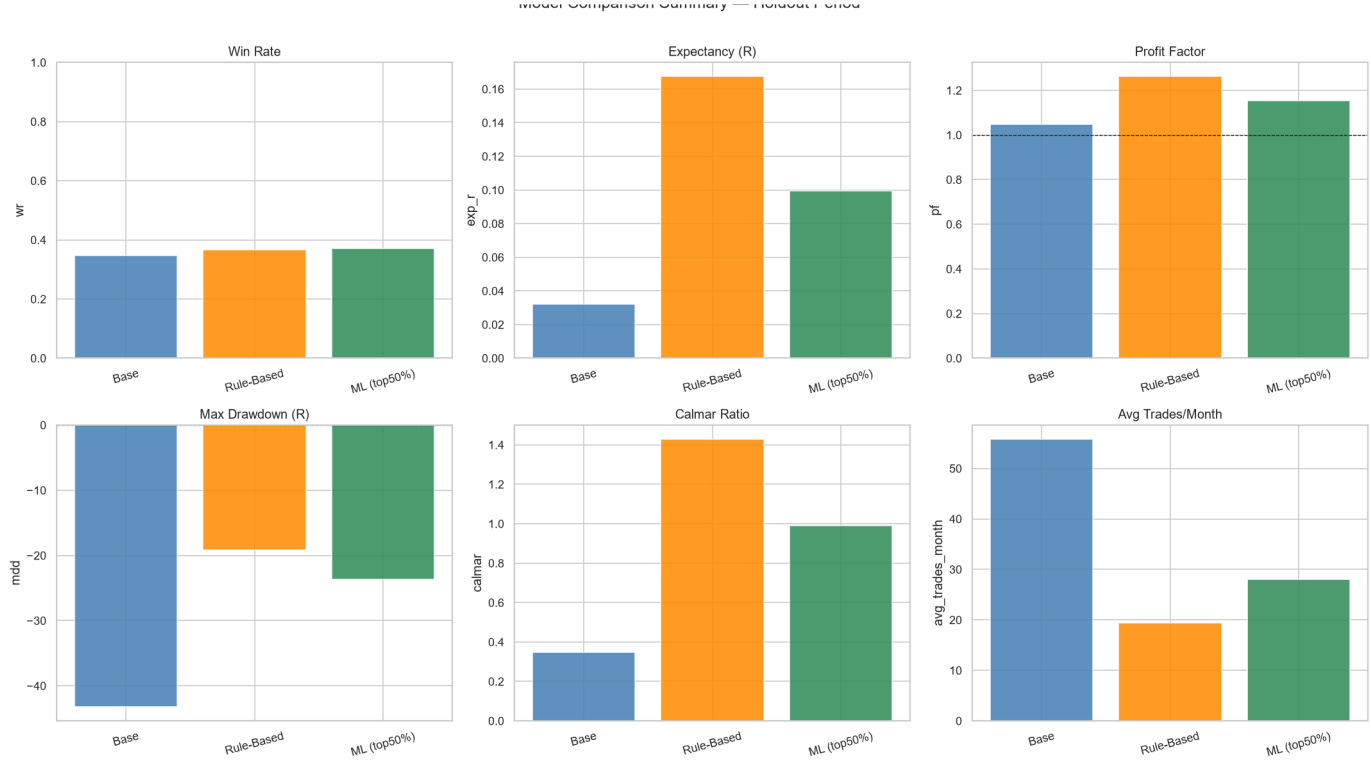
Criteria:

1. Bootstrap 95% CI lower bound > 0
2. Profit factor > 1.1
3. Max drawdown > −15 R

4. Avg trades/month ≥ 20
5. Tiebreak: highest Calmar ratio

Model	CI_lo > 0	PF > 1.1	MDD > −15R	Trades ≥ 20/mo	Passes All	Calmar
Base	✗	✗	✗	✓	No	0.347
Rule-Based	✗	✓	✗	✗	No	1.429
ML (top 50%)	✗	✓	✗	✓	No	0.989

> **CAUTION**
> **No model passes all pre-specified criteria on the holdout period.** The notebook output states: "NO MODEL passes all criteria on holdout. Review thresholds or extend data."



Summary Dashboard

Threats to Validity

1. **Single instrument** — All results are for XAUUSDm only. Performance may not generalise to other instruments or markets.
2. **Backtest bias** — The 4-year and 5-year files were generated by the same system; survivorship bias and look-ahead in strategy design cannot be fully ruled out.
3. **Parameter tuning** — 20 rule variants were tested in EDA 2. With Bonferroni-corrected $\alpha = 0.0025$, any finding with $p < 0.001$ should be treated as potentially spurious without independent confirmation.
4. **Small holdout sample** — The holdout contains 549 base trades, 191 rule-based trades, and 275 ML trades (≈10 months). Bootstrap confidence intervals span 0 across all models, and no pairwise comparison is statistically significant after BH correction.
5. **Concurrent open positions** — 69.8% of 4-yr entries and 54.8% of 5-yr entries had another position open simultaneously, limiting the effective independent sample size.
6. **ML signal marginality** — XGBoost achieved validation AUC = 0.5011 (permutation p = 0.5000) and holdout AUC = 0.5168. While top 50% scoring improved expectancy (+0.0995 R vs +0.0320 R Base), the observed difference (+0.067 R) is not statistically distinguishable from chance (BH p = 0.3220).
7. **ATR regime shift** — `atr_h4_pips` mean is 1,213 in train vs 4,437 in holdout, indicating the holdout period coincided with severe volatility expansion.

Report compiled from outputs of notebooks `00_data_audit.ipynb` through `04_eda_comparison.ipynb`. All values are taken directly from cell outputs; no values have been estimated or assumed.